

a Continuous fragment growth

01 Extend the frontier

Choose a strong source edge.
Check all active fragment bonds.
Carry compatible placements.

02 Preserve distinctions

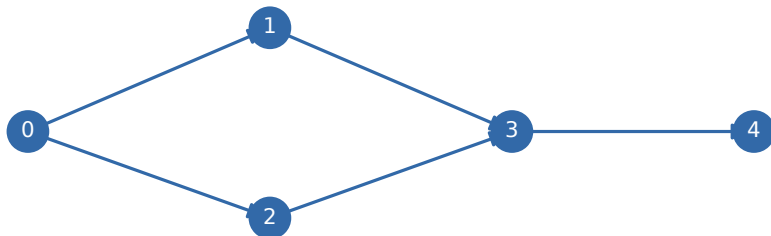
Canonicalize candidate states.
Keep locked roles and boundaries.
Retain correlated symmetry.

03 Continue or branch

Grow until saturation.
Fork distinct fragment placements.
Continue under each prefix.

repeat after each accepted extension

b Shared search history



Fork

Exact state join

04 Compile retained families

Merge equal or contained families.
Keep incomparable alternatives.
Preserve every discovering path.

Schematic: joins require equal assignments, island partitions and deferred edges at a compatible continuation.